

past paper 2016

(Short Questions)

Q2(a)

prove that $y = d_1 e^{(a+ib)x} + d_2 e^{(a-ib)x}$ is a algebraic equivalent to $y = C_1 e^{ax} \cos bx + C_2 e^{ax} \sin bx$

soln: $y = d_1 e^{ax} e^{ibx} + d_2 e^{ax} e^{-ibx}$
 $= e^{ax} [d_1 e^{ibx} + d_2 e^{-ibx}]$
 $= e^{ax} [d_1 \cos bx + d_1 i \sin bx + d_2 \cos bx - d_2 i \sin bx]$

$y = e^{ax} [(d_1 + d_2) \cos bx + (d_1 i - d_2 i) \sin bx] \text{ (Ans.)}$

Q2(b)

Define Cauchy Euler's equation.

Ans: Cauchy Euler's with n th term

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_2 x^2 \frac{d^2 y}{dx^2} +$$

$$a_1 x \frac{dy}{dx} + a_0 y = f(x)$$

Example :-

$$7x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} + 7y = \sin(\ln x)$$

Q2(c)

Find particular integral of

$$(D^3 - D^2 - D + 1)y = e^x$$

soln:

For

y_p

$$y_p = \frac{1}{D^3 - D^2 - D + 1} e^x$$

$$= \frac{x}{3D^2 - 2D - 1} e^x$$

$$= \frac{x^2}{6D - 2} e^x$$

$$= \frac{x^2}{6(1) - 2} e^x$$

$$y_p = \frac{x^2}{4} e^x$$

Q2(d)

Convert $y' = 2xy - x$ in exact differential equation.

soln:

$$y' = 2xy - x$$

$$y' - 2xy = -x \quad \text{--- (i)}$$

$$\int -2x dx = -x^2$$

$$I.F = e^{-x^2}$$

Multiply e^{-x^2} by eq (i)

$$e^{-x^2} dy - 2xy e^{-x^2} dx = -x e^{-x^2} dx$$

$$(x e^{-x^2} - 2xy e^{-x^2}) dx + e^{-x^2} dy = 0$$

Q2 (e)

Find the solution of the following differential equation.

$$xy' - 2x^2y = y \ln y$$

soln: $\frac{x}{y \ln y} y' - \frac{2x^2}{\ln y} = 1$

put $\ln y = u$

Diff w.r.t 'x'

$$\frac{1}{y} \frac{dy}{dx} = \frac{du}{dx}$$

$$\frac{x}{u} \frac{du}{dx} - \frac{2x^2}{u} = 1$$

$$x \frac{du}{dx} - 2x^2 = u \quad \text{--- (i)}$$

$$\frac{du}{dx} - \frac{u}{x} = 2x$$

$$I.F = e^{\int -\frac{1}{x} dx} = e^{-\ln x}$$

$$I.F = e^{\ln x^{-1}} = \frac{1}{x}$$

Multiply $\frac{1}{x}$ by eq (i)

$$\frac{1}{x} \frac{du}{dx} - \frac{u}{x^2} = 2$$

$$\frac{d}{dx} \left(\frac{u}{x} \right) = 2x$$

Integrate both sides

$$\frac{u}{x} = 2x + C$$

$$u = 2x^2 + Cx$$

$$\ln y = \frac{2x^2 + Cx}{2x^2 + Cx}$$

$$y = e^{2x^2 + Cx}$$

$$y = e^{Cx} \cdot e^{2x^2}$$

$$y = C_1 e^{2x^2}$$

$$\therefore e^{Cx} = C_1$$

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past paper 2015

(Short Questions)

Q2 (i) Ans:

$$\frac{dy}{dx} = -x \text{ with } y(3) = 4$$

soln: $dy = -x dx$
Integrate both sides

$$y = -\frac{x^2}{2} + C \quad \text{--- (i)}$$

for condition

$$y(3) = 4$$

$$x = 3, y = 4$$

put in (i)

$$4 = -\frac{(3)^2}{2} + C$$

$$C = 4 + \frac{9}{2}$$

$$C = \frac{17}{2}$$

$$\text{so, } y = -\frac{x^2}{2} + \frac{17}{2}$$

Q2 (ii) Ans.

Cauchy-Euler equation:

An equation of the form

$$\alpha_0 x^n \frac{d^n y}{dx^n} + \alpha_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + \alpha_{n-1} x \frac{dy}{dx} + \alpha_n y = f(x)$$

is called Cauchy-Euler equation
 $\alpha_0, \alpha_1, \dots, \alpha_{n-1}, \alpha_n$ are real constants.

Example:-

$$x^4 \frac{d^4 y}{dx^4} + 6x^3 \frac{d^3 y}{dx^3} + 9x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + y = (1 + \ln x)^2$$

Q2 (iii) Ans.

$$y = \cos x - e^{-x}$$

soln: Diff w.r.t 'x'

$$\frac{dy}{dx} = -\sin x + e^{-x}$$

$$\Rightarrow \frac{dy}{dx} + \sin x = e^{-x}$$

which is differential equation

Q2 (iv) Ans:

$$\frac{dy}{dx} = \frac{2y-x+5}{2x-y-4}$$

soln: put $x = x+h$

$$y = y+k$$

where h, k are constants

so, $dy = dy$, $dx = dx$

$$\begin{aligned}\frac{dy}{dx} &= \frac{2(y+k)-(x+h)+5}{2(x+h)-(y+k)-4} \\ &= \frac{2y+2k-x-h+5}{2x+2h-y-k-4}\end{aligned}$$

$$\frac{dy}{dx} = \frac{2y-x+2k-h+5}{2x-y+2h-k-4} \quad \text{--- (i)}$$

Now put $2k-h+5=0$ --- (ii)

$$2h-k-4=0 \quad \text{--- (iii)}$$

Multiply 2 by (ii) and add (i)

$$4k-2h+10=0$$

$$2h-k-4=0$$

$$3k+6=0$$

$$k = -2$$

put in (ii)

$$2h+2-4=0$$

$$2h=2 \Rightarrow h=1$$

so, $\frac{dy}{dx} = \frac{2y-x}{2x-y}$ which is homogenous equation

Q2 (v) Ans:

$$(x^2 - 2x + 2y^2)dx + 2xy dy = 0$$

soln:

$$M = x^2 - 2x + 2y^2, \quad N = 2xy$$

$$M_y = 4y, \quad N_x = 2y$$

so,

$$\frac{M_y - N_x}{N} = p(x)$$

$$\frac{4y - 2y}{2xy} = p(x)$$

$$p(x) = \frac{2y}{2xy} = \frac{1}{x}$$

$$I.F = e^{\int p(x) dx}$$

$$= e^{\int \frac{1}{x} dx}$$

$$I.F = e^{\ln x}$$

$$I.F = x$$

Q2(vi) Ans:

$$\frac{d^2y}{dx^2} - 4y = e^x + \sin x$$

$$\text{soln: } (D^2 - 4)y = e^x + \sin x$$

For y_c

$$(D^2 - 4)y = 0$$

Auxiliary eq is

$$m^2 - 4 = 0$$

$$m = \pm 2$$

$$y_c = (C_1 e^{2x} + C_2 e^{-2x}) \text{ Ans}$$

Q2 (viii) Ans:

$$(x^2 - 5x + 6) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 4y = 0$$

soln: Divide $(x^2 - 5x + 6)$ on B.S

$$\frac{d^2 y}{dx^2} - \frac{x}{(x^2 - 5x + 6)} \frac{dy}{dx} + \frac{4}{(x^2 - 5x + 6)} y = 0$$

compare it with

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + Q(x) y = 0$$

$$p(x) = -\frac{x}{x^2 - 5x + 6}, \quad Q(x) = \frac{4}{x^2 - 5x + 6}$$

For singular point

$$x^2 - 5x + 6 = 0$$

$$x^2 - 3x - 2x + 6 = 0$$

$$(x - 3)(x - 2) = 0$$

$$x = 3, 2$$

There are singular points

But ~~ord~~ ordinary points are

$$\begin{aligned} \text{Ordinary point} &= \mathbb{R} - \text{singular point} \\ &= \mathbb{R} - \{2, 3\} \end{aligned}$$

Q2 (ix) Ans:

Let p = population
so rate increasing
population is $\frac{dp}{dt}$

Given: $\frac{dp}{dt} \propto p$

$$\frac{dp}{dt} = kp$$

where k is proportionality
constant.

$$\frac{dp}{dt} - kp = 0$$

which is a differential
equation.

Long Questions

Q3. Ans:

$$\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4y = 0$$

$$\text{soln: } (D^3 - 3D^2 + 4)y = 0$$

Auxiliary eq is

$$m^3 - 3m^2 + 4 = 0$$

solve by using synthetic division

$$\begin{array}{r|rrrrr} -1 & 1 & -3 & 0 & 4 \\ & \downarrow & -1 & 4 & -4 \\ \hline & 1 & -4 & 4 & 0 \end{array}$$

$m = -1$ are one factor

Depressed eq. is

$$m^2 - 4m + 4 = 0$$

$$m^2 - 2m - 2m + 4 = 0$$

$$m(m-2) - 2(m-2) = 0$$

$$(m-2)(m-2) = 0$$

$$m = 2, 2$$

$$y_c = C_1 e^{-x} + (C_2 + C_3 x) e^{2x}$$

$$y = C_1 e^{-x} + C_2 e^{2x} + C_3 x e^{2x} \quad \dots (i)$$

for condition

$$y(0) = 1$$

$$x=0, y=1$$

$$1 = C_1 e^0 + C_2 e^0 + 0$$

$$C_1 + C_2 = 1 \quad \dots (ii)$$

~~for condition~~

$$y' = -C_1 e^{-x} + 2C_2 e^{2x} + C_3 [e^{2x} + 2x e^{2x}]$$

$$y' = -C_1 e^{-x} + 2C_2 e^{2x} + C_3 e^{2x} + 2xC_3 e^{2x}$$

for condition

$$y'(0) = -8$$

$$x=0, y = -8$$

$$-8 = -C_1 e^0 + 2C_2 e^0 + C_3 e^0 + 0$$

$$-8 = -C_1 + 2C_2 + C_3$$

$$C_3 + 2C_2 - C_1 = -8 \text{ --- (iii)}$$

$$y'' = C_1 e^{-x} + 4C_2 e^{2x} + 2C_3 e^{2x} + 2C_3 [e^{2x} + 2xe^{2x}]$$

$$= C_1 e^{-x} + 4C_2 e^{2x} + 2C_3 e^{2x} +$$

$$2C_3 e^{2x} + 4xC_3 e^{2x}$$

$$y'' = C_1 e^{-x} + 4C_2 e^{2x} + 4C_3 e^{2x} + 4xC_3 e^{2x}$$

for condition

$$y''(0) = -4$$

$$x=0, y = -4$$

$$-4 = C_1 e^0 + 4C_2 e^0 + 4C_3 e^0 + 0$$

$$-4 = C_1 + 4C_2 + 4C_3$$

$$C_1 + 4C_2 + 4C_3 = -4 \text{ --- (iv)}$$

~~Add~~ Multiply 4 by eq (iii) and subtract from (iv)

$$4C_3 + 8C_2 - 4C_1 = -32$$

$$+ C_1 + 4C_2 + 4C_3 = -4$$

$$-5C_1 + 4C_2 = -28 \text{ --- (v)}$$

Multiply 5 by eq (ii) and add in (v)

(3)

$$\begin{aligned}
 5C_1 + 5C_2 &= 5 \\
 -5C_1 + 8C_2 &= -28 \\
 \hline
 9C_2 &= 3 - 23 \\
 C_2 &= -\frac{20}{9}
 \end{aligned}$$

put in (ii)

$$C_1 - \frac{20}{9} = 1$$

$$C_1 = 1 + \frac{20}{9}$$

$$C_1 = \frac{32}{9}$$

put in (iii)

$$C_3 + 2\left(-\frac{20}{9}\right) - \frac{32}{9} = -8$$

$$C_3 - \frac{40 - 32}{9} = -8$$

$$C_3 - \frac{8}{9} = -8$$

$$C_3 = -8 + \frac{8}{9} = -\frac{72}{9} + \frac{8}{9}$$

$$C_3 = -\frac{64}{9}$$

put values in (i)

$$y = \frac{32}{9}e^{-x} - \frac{20}{9}e^{2x} + \frac{2x}{3}e^{2x} \quad \text{Ans.}$$

Q4. Ans:

$$2\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = x^2 + 3\sin x \text{ --- (i)}$$

soln: $(2D^2 + 3D + 1)y = x^2 + 3\sin x$

For y_c

$$(2D^2 + 3D + 1)y = 0$$

Auxiliary eq is

$$2m^2 + 3m + 1 = 0$$

~~can~~ solve by using quadratic formula

$$m = \frac{-3 \pm \sqrt{9-8}}{2(2)}$$

$$= \frac{-3 \pm \sqrt{1}}{4} \Rightarrow \frac{-3 \pm 1}{4}$$

$$m = \frac{-3+1}{4}, \frac{-3-1}{4}$$

$$m = -\frac{1}{2}, -1$$

$$y_c = C_1 e^{(-1/2)x} + C_2 e^{-x}$$

For particular soln of eq (i)

$$2y'' + 3y' + y = x^2 \text{ --- (ii)}$$

$$2y'' + 3y' + y = 3\sin x \text{ --- (iii)}$$

from eq (ii)

$$y_p = Ax^2 + Bx + C$$

$$y_p' = 2Ax + B$$

$$y_p'' = 2A$$

put in (ii)

$$2(2A) + 3(2Ax + B) + Ax^2 + Bx + C = x^2$$

$$4A + 6Ax + 3B + Ax^2 + Bx + C = x^2$$

$$Ax^2 + (6A + B)x + (4A + 3B + C) = x^2$$

Equating coeff of x^2, x, const

$$A = 1$$

$$6A + B = 0$$

$$B = -6$$

$$4A + 3B + C = 0$$

$$4(1) + 3(-6) + C = 0$$

$$-14 + C = 0$$

$$C = 14$$

$$y_p = x^2 - 6x + 14$$

From eq (iii)

$$y_p = (E \sin x + F \cos x) x^k$$

$$y_p = E \sin x + F \cos x$$

$$y_p' = E \cos x - F \sin x$$

$$y_p'' = -E \sin x - F \cos x$$

put in eq (iii)

$$2(-E \sin x - F \cos x) + 3(E \cos x - F \sin x)$$

$$+ E \sin x + F \cos x = 3 \sin x$$

$$-2E \sin x - 2F \cos x + 3E \cos x - 3F \sin x$$

$$+ E \sin x + F \cos x - 3 \sin x$$

$$- E \sin x - F \cos x + 3 E \cos x - 3 F \sin x$$

$$= 3 \sin x$$

$$(-3F - E) \sin x + (3E - F) \cos x = 3 \sin x$$

Equating coefficient of
 $\sin x, \cos x$

$$-3F - E = 3 \rightarrow 3F + E = -3 \quad \text{--- (A)}$$

$$3E - F = 0 \quad \text{--- (B)}$$

Multiply 3 by (B) and add
in (A)

$$9E - 3F = 0$$

$$3F + E = -3$$

$$10E = -3$$

$$E = \frac{-3}{10}$$

put in (A)

$$3F + \frac{-3}{10} = -3$$

$$3F = -3 + \frac{3}{10}$$

$$3F = \frac{-27}{10}$$

$$F = \frac{-9}{10}$$

put in (iii)

$$y_p = \frac{3}{10} \sin x - \frac{9}{10} \cos x$$

$$y = (C_1 e^{(10)x} + C_2 e^{-x}) + (x^2 - 6x + 14) + \frac{3}{10} \sin x + \frac{9}{10} \cos x$$

Q5. Ans.

$$\frac{dy}{dx} + \frac{xy}{1-x^2} = xy^{1/2} \quad \text{--- (i)}$$

soln:

Divide by $y^{1/2}$ eq (i)

$$y^{-1/2} \frac{dy}{dx} + \frac{xy}{1-x^2} y^{-1/2} = x \quad \text{--- (ii)}$$

$$y^{-1/2} \frac{dy}{dx} + \frac{xy^{1/2}}{1-x^2} = x$$

$$y^{-1/2} \frac{dy}{dx} + \frac{x}{1-x^2} y^{1/2} = x \quad \text{--- (ii')}$$

put $y^{1/2} = v$
Diff w.r.t x

$$\frac{1}{2} y^{-1/2} \frac{dy}{dx} = \frac{dv}{dx}$$

$$y^{-1/2} \frac{dy}{dx} = 2 \frac{dv}{dx}$$

put in (ii')

$$2 \frac{dv}{dx} + \frac{x}{1-x^2} v = x$$

$$\frac{dv}{dx} + \frac{x}{2(1-x^2)} V = \frac{x}{2} \quad \text{--- (iii)}$$

Which is linear diff eq in V
 I.f = $e^{\int \frac{x}{2(1-x^2)} dx}$

$$\begin{aligned} &= e^{\frac{1}{2} \int \frac{x}{1-x^2} dx} \\ &= e^{-\frac{1}{4} \int \frac{-2x}{1-x^2} dx} \\ &= e^{-\frac{1}{4} \ln(1-x^2)} \\ &= e^{\ln(1-x^2)^{-1/4}} \\ &= (1-x^2)^{-1/4} \\ \text{I.f} &= \frac{1}{(1-x^2)^{1/4}} \end{aligned}$$

Multiply by eq (iii)

$$\frac{1}{(1-x^2)^{1/4}} \frac{dv}{dx} + \frac{x}{2(1-x^2)^{5/4}} V = \frac{x}{2(1-x^2)^{1/4}}$$

$$\frac{d}{dx} \left[\frac{1}{(1-x^2)^{1/4}} V \right] = \frac{x}{2(1-x^2)^{1/4}}$$

Integrate both sides

$$\begin{aligned} \frac{V}{(1-x^2)^{1/4}} &= \frac{1}{2} \int \frac{x}{(1-x^2)^{1/4}} dx \\ &= -\frac{1}{4} \int -2x(1-x^2)^{-1/4} dx \end{aligned}$$

$$\begin{aligned} \frac{V}{(1-x^2)^{1/4}} &= -\frac{1}{4} \left[\frac{(1-x^2)^{-1/4+1}}{-1/4+1} \right] + C \\ &= -\frac{1}{4} \left[\frac{(1-x^2)^{3/4}}{3/4} \right] + C \end{aligned}$$

$$\frac{V}{(1-x^2)^{1/4}} = -\frac{1}{3} (1-x^2)^{3/4} + C$$

$$V = C(1-x^2)^{1/4} - \frac{1}{3} (1-x^2)$$

put $y^{1/2} = V$

$$y^{1/2} = C(1-x^2)^{1/4} - \frac{(1-x^2)}{3} \text{ Ans.}$$